

# GameSparcs: Enhancing development efficiency and offering smooth gaming experiences with containerized infrastructure

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GameSparcs (TWSE: 6542) is a global game platform operation company founded in 2010 that has been diversifying its business over the recent years. Besides mobile game distribution, GameSparcs also develops online social game platforms and licenses LAN-based gaming systems. By integrating enterprise resources, diversifying its business and expanding to overseas markets, GameSparcs aims at enhancing its business competitiveness.

Industries: Games

By using the containerized cloud infrastructure on Google Cloud to develop games and development products, GameS

ensure smooth g  
services while en  
DevOps team's ef

Location: Taiwan

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

## Google Cloud results

- Enable high software portability across different operating environments with GKE
- Support one time faster feature deployment compared to VMs with container instances in GKE
- Accelerate game loading speed by two times with Cloud Load Balancing

Support  
smooth gaming  
experiences  
during traffic  
surges

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

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Online games are a major source of entertainment for many people, with seemingly endless games available on the market. Facing fierce competition, game developers need to ensure excellent gaming experiences while constantly releasing new products to attract and retain users.

Following the market trend, GameSparcs (<https://www.gamesparcs.com/en>) has been diversifying its business over the recent years. Starting as a game publisher in Taiwan by localizing popular mobile games from China, Japan, and Korea, the company has gradually expanded its portfolio to social game platform and LAN-based gaming system development. To offer better gaming experiences, GameSparcs is constantly developing new products, with at least one new game added to its game platforms every month. Currently, the game developer offers premium gaming experiences to all types of gamers across Europe, America, and Asia through three online social game platforms, including Bravo Casino, a popular interactive social game platform; Ace Club, which features diverse gameplay; and Jackpot Island, which is designed for the European and American markets.

"We want to provide diverse gaming experiences that meet users' needs and enhance our competitiveness," says Howard Liao, Information Technology Director at GameSparcs. "Besides universally popular casino games like poker, slot machines, and fishing, we've also developed games adapted to different cultural contexts, so as to cater to gamers' wide-ranging preferences."

To reduce operational workloads and allow its DevOps team to focus on software development, GameSparcs decided to adopt a public cloud platform that supports automated scalability and simple infrastructure management when developing its own games. The company eventually chose Google Cloud as it has a large number of data centers and network nodes around the world, offering speedy, reliable and low-latency connectivity that can enable quality gaming experiences, on top of meeting GameSparcs' requirements for cloud infrastructure.

"We're constantly developing and releasing new products, so it's important for us to run our games in a cloud environment that enables quick development and high development

operational  
efficiency.  
Google Cloud  
not only ticks  
this box but also  
supports low  
latency with its  
extensive global  
network."

*—Howard Liao,  
Information Technology  
Director, GameSparcs*

## **Enhancing DevOps efficiency while ensuring service stability with GKE**

GameSparcs has been using container instances in  
[Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine)

(<https://cloud.google.com/kubernetes-engine>) (GKE) to  
run its three social game platforms since their  
launch. Howard Liao points out that the  
containerized architecture has greatly simplified the  
team's DevOps tasks by clearly separating the  
development and operational workloads, and  
enabling high software portability. This means that  
GameSparcs' games can be easily deployed in  
different types of host environments without any  
adjustments. The operation of each game can also  
be monitored separately on the GKE dashboard,

making it easier for the team to manage its services and troubleshoot. Overall, GameSparcs has not only realized highly efficient operations, but also significantly enhanced effectiveness. Compared to traditional operations methods, the effectiveness of GameSparcs' operations using GKE is 40% higher, which supports smoother gaming experiences. At the same time, it has optimized its labor costs by reducing the staff needed for the deployment by 30%, enabling support for the operations of hundreds of games.

The autoscaling feature of GKE has also enabled GameSparcs to effortlessly offer high availability. The traffic of its social game platforms usually fluctuates instantly when the company runs a marketing campaign. By automatically adjusting the number of its container instances to meet the current needs, GKE has made it possible for GameSparcs to use the necessary resources without any manual intervention. According to the team's comparison test before adopting GKE, it would have required two to three days of work to manually scale on virtual machines (VMs) at the same level.

"We adopt  
containerized  
architecture for  
all our games to

enhance our  
DevOps  
efficiency. By  
deploying  
container  
instances in  
GKE, we're able  
to fully enjoy the  
benefits of  
containerization,  
realize high and  
instant  
scalability with  
minimum effort,  
and establish  
more  
comprehensive  
container  
management  
strategies. Users  
of our game  
platforms have  
never  
experienced any  
disruptions due  
to instant traffic  
increase."

*—Howard Liao,  
Information Technology  
Director, GameSparcs*

## **Supporting high service availability and low network latency with Cloud Load Balancing**

To further secure the availability of its gaming services, GameSparcs employs Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to connect end users to the instances that host its games in GKE. By grouping instances to traffic, Cloud Load Balancing helps prevent a full-scale downtime when one instance is out of service. As a result, its games have experienced zero outages since the adoption of Cloud Load Balancing.

Since Cloud Load Balancing can connect gamers to its game servers from the nearest network node, the extensive global network of Google Cloud has enabled GameSparcs to realize faster game loading speed for gamers around the world. Liao notes the company's comparison test shows that the content loading speed through Google Cloud is at least two times faster than other cloud platforms.

"Service availability and connectivity consist a big part of the gaming experiences that we offer, because casino games are highly interactive. With Cloud Load Balancing supporting high availability and speedy connectivity, our users can fully enjoy the excitement of our games," he adds.



## Optimizing infrastructure management and delivering more engaging gaming experiences with AI

Following this, GameSparcs will transform the architecture of its games into service meshes in GKE to further simplify management and enhance portability, while leveraging BigQuery.

(<https://cloud.google.com/bigquery>) to store users' history records to process data queries quickly. The team also plans to use the generative artificial intelligence (AI) technology of Google to turn written game guides into video presentations with virtual humans, increasing its games' interactivity.

"By providing a powerful cloud infrastructure and tools that are easy-to-use, diverse, widely applicable, highly efficient, stable and secure, as well as easy to scale and redevelop,

Google Cloud  
has enabled us  
to offer excellent  
gaming  
experiences and  
release new  
products  
frequently with  
higher DevOps  
efficiency. We'll  
continue  
optimizing our  
operations and  
developing  
innovative  
features to  
upgrade our  
services."

—Howard Liao,  
Information Technology  
Director, GameSparcs